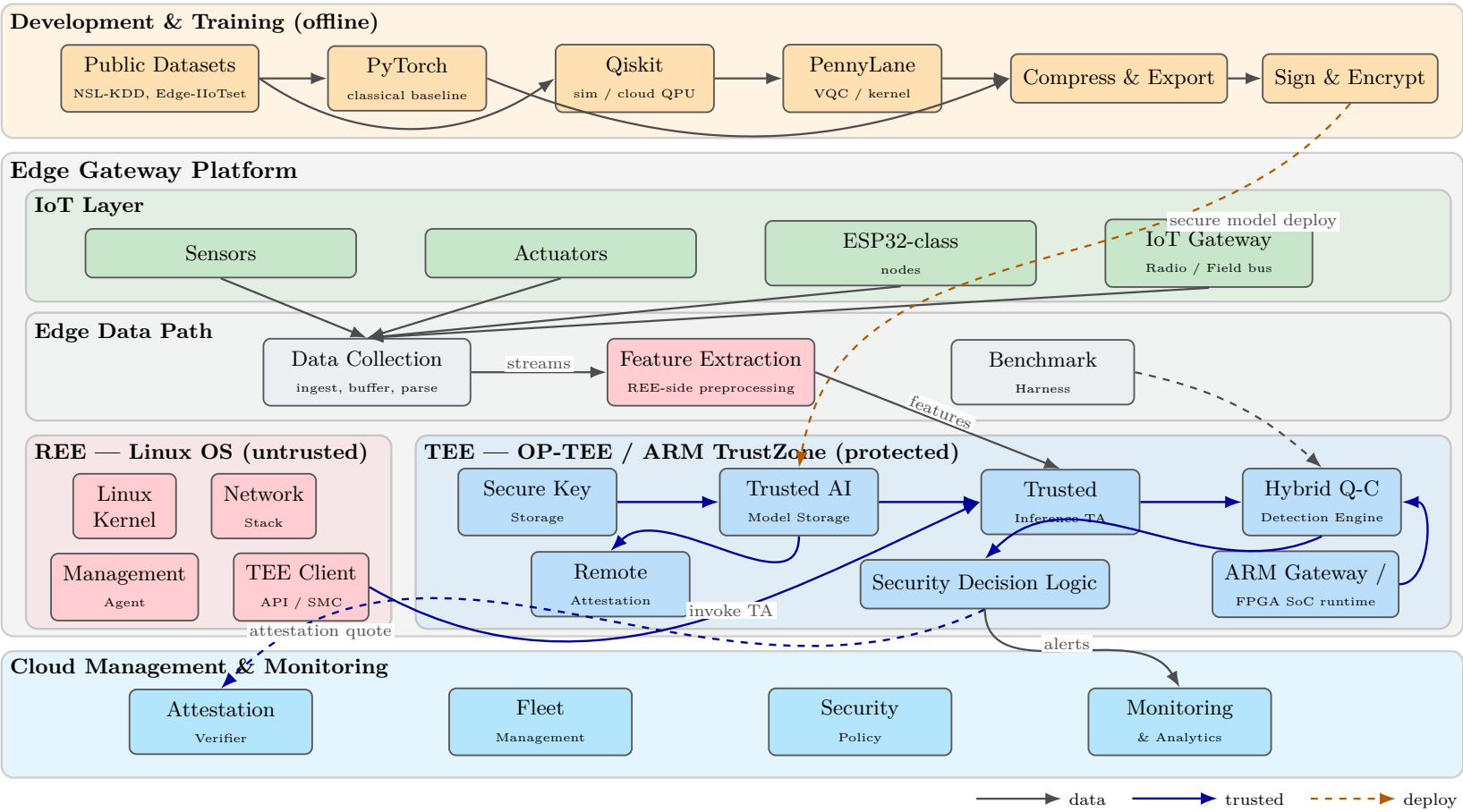


Trustworthy Hybrid Quantum–Classical AI for Secure IoT Edge Systems

System Architecture — TEE-protected deployment, trusted inference, and hybrid anomaly detection



RQ:
Hybrid Q-C vs. TinyML for zero-day / unseen attacks under deployability & TEE constraints

Threats mitigated:
model tampering; unauthorized replacement; extraction; unauthorized inference; compromised REE

TEE protects:
model storage, inference, keys, decision logic

QAI role:
anomaly engine inside TEE; not a security primitive

Flow and methodology:

- Offline training:** public datasets are used to train classical (PyTorch) and quantum (Qiskit/PennyLane) models; outputs are compressed, signed, and encrypted before deployment to trusted model storage in the TEE.
- Edge data path:** sensors, actuators, and field nodes send telemetry to the gateway; data is ingested, buffered, and parsed on the edge, with feature extraction performed in the untrusted REE.
- Trusted inference:** the TEE Client API invokes the inference trusted application, which loads the protected model and runs hybrid quantum–classical anomaly detection inside OP-TEE / ARM TrustZone.
- Security and monitoring:** attestation quotes and security alerts are sent to the cloud verifier and monitoring services for integrity checks, fleet policy enforcement, and incident response.

Research questions:

RQ1. Why is TinyML insufficient, and why is Quantum AI necessary?

- TinyML deploys compressed *classical* models on constrained edge hardware, but decision boundaries are learned only from known attack patterns; performance drops on **zero-day**, **stealthy**, and **distribution-shift** traffic typical of IIoT/IoT security datasets (e.g. NSL-KDD, Edge-IIoTset).
- Aggressive quantization and pruning under fixed power P and latency T budgets further limits representational capacity—classical TinyML often trades detection recall for deployability before meeting accuracy targets.
- High-dimensional telemetry features admit **adversarial evasion** against shallow/quantized classical models; a stronger detector is needed without abandoning edge constraints.
- Hybrid quantum–classical models (VQC, quantum kernels, quantum-inspired compression) provide a **testable alternative**: richer feature maps at comparable model size, benchmarked head-to-head against the PyTorch TinyML baseline via the edge harness in this architecture.

RQ2. What is the role of Quantum AI? What is the role of the TEE?

- Role of Quantum AI:** the hybrid Q–C detection engine *inside* the TEE performs anomaly/intrusion scoring on REE-extracted features; it is **not a security primitive** but an AI workload whose outputs feed security decision logic. Its purpose is improved detection under deployability constraints; negative results remain valid evidence.
- Role of the TEE:** OP-TEE / ARM TrustZone provides a hardware-backed trust boundary that isolates secure key storage, trusted model storage, inference TA, remote attestation, and decision logic from the compromised Linux REE. It enables signed model deploy, trusted TA invocation, and verifiable attestation quotes to the cloud.
- Division of responsibility:** **QAI addresses detection**; **TEE addresses protection** of models, keys, inference runtime, and integrity evidence (model tampering, unauthorized replacement/extraction/inference, REE compromise).